

- **Sentiment Analysis**

Sentiment Analysis is the process of identifying whether a customer's opinion is **positive, negative, or neutral** by analyzing their reviews or feedback.

- Types

There are three types of sentiments **Positive, Negative, and Neutral**.

- Applications

It is commonly used in e-commerce websites, social media platforms, and product review systems to understand customer opinions.

- Advantages

It helps companies know what customers like or dislike about their products. This allows businesses to improve their products and provide better customer service.

- **NLP (Natural Language Processing)**

Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that helps computers understand, analyze, and process human language.

- In NLP, the text is first collected, then cleaned, processed, analyzed, and finally classified to understand its meaning.

- Real-life applications

It is used in applications like ChatGPT, Google Translate, voice assistants, spam detection, and chatbots

- Importance in sentiment analysis

In sentiment analysis, NLP helps the computer identify whether a review expresses a positive, negative, or neutral opinion.

- **Dataset Understanding**

The dataset is stored in a CSV (Comma-Separated Values) file named **clean_amazon_reviews_1.csv**. It contains customer reviews of Amazon products and is used to perform sentiment analysis by examining customer opinions.

Explain Every Column

Column	Description
name	Name of the product.
brand	Brand or company of the product.
reviews.date	Date when the customer posted the review.
reviews.rating	Rating given by the customer (1–5 stars).
reviews.text	Complete review written by the customer.
reviews.title	Short title of the review.
reviews.doRecommend	Indicates whether the customer recommends the product (TRUE/FALSE).
reviews.numHelpful	Number of users who found the review helpful.
clean_review	Cleaned version of the review used for text analysis.
Sentiment	Sentiment label of the review (Positive, Negative, or Neutral).

Number of Rows and Columns

- **Total Rows: 34,624**
- **Total Columns: 10**

This means the dataset contains **34,624 customer reviews** with **10 different attributes** describing each review.

Missing Values

The dataset is almost complete.

- **No missing values** in 9 columns.
- **Only 6 missing values** in the **reviews.title** column.
- Since only a few titles are missing, they have very little impact on the overall analysis.

- **Dataset Summary**

The dataset contains **34,624 customer reviews** collected from Amazon products. It includes information such as the product name, brand, review date, customer rating, review title, complete review text, recommendation status, helpful vote count, cleaned review text, and sentiment label. The reviews are classified into **Positive, Negative, and Neutral** sentiments, making the dataset suitable for sentiment analysis.

The **reviews.rating** column records customer ratings from **1 to 5 stars**, helping measure customer satisfaction. The **reviews.text** column stores the original customer feedback, while the **clean_review** column contains a processed version of the text that is easier for analysis. The **reviews.doRecommend** column indicates whether the customer recommends the product, and **reviews.numHelpful** shows how many users found the review useful.

The dataset has **10 columns and 34,624 rows**, providing a large amount of customer feedback for analysis. It contains **very few missing values**, with only **6 missing entries in the reviews.title column**, indicating that the dataset is clean and reliable. Overall, this dataset is well-suited for analyzing customer opinions, identifying sentiment patterns, generating visualizations, and helping businesses understand customer satisfaction and improve their products.

- **Buyer Journey**

Awareness

At this stage, the customer becomes aware of a product through advertisements, social media, websites, or recommendations from others.

Consideration

The customer compares different products by checking features, prices, ratings, and customer reviews before making a decision.

Purchase

After comparing the available options, the customer selects the product and completes the purchase.

Post-Purchase

After using the product, the customer shares their experience by giving ratings and writing reviews, which help other buyers.

How Sentiment Analysis Helps Buyers

Sentiment analysis examines customer reviews and identifies whether they are positive, negative, or neutral. It helps buyers quickly understand public opinion about a product, making it easier to choose the right product and avoid poor-quality ones.

